

What pitch should Tarik Skubal throw?

Relevant training course: Data Scientist

Level of difficulty: 9/10

Description of the project:

Determine the *ideal pitch* that a pitcher (for example, Tarik Skubal, Garrett Crochet... or maybe a lesser-known pitcher!) should throw to **maximize their chances of success**, while taking into account the precise context of the game: score, count, batter, inning, etc.



Data & Tool

To do this, we'll use a real gem from the Python ecosystem: the **pybaseball** library!

 Official documentation → [pybaseball · PyPI](https://pybaseball.readthedocs.io/en/latest/)

This library allows us to:

- ✓ Extract highly detailed data on **all MLB games** (data updated daily)
- ✓ Get statistics by player, team, or match
- ✓ Access millions of pitches with **exceptional granularity**: pitch type, velocity, spin, outcome, game situation, and more

The goal is to answer the following question:



"Which pitch, in which situation, truly maximizes the chance of success?"

In the end, we aim to build a model that could even help a pitcher decide in real time whether to throw a slider, fastball, or changeup to strike out the batter. ⚾

(You can also start with simpler tasks, like predicting the pitch type, the outcome of the pitch, or the winning team.)

Conditions for project validation :

- a data exploration, data visualization and pre-processing **report** ;
- a modeling **report** ;
- a **final report** and associated **GitHub**.